Clinical Guideline for:
Venous Thromboembolism (VTE) Prophylaxis on ICU

Summary

- ALL ICU admissions (even elective surgical) are high risk for VTE and require some form of prophylaxis.
- Most patients can be prescribed pharmacological prophylaxis (see below)
- ALL patients should have non-pharmacological thromboprophylaxis (graduated compression stockings OR foot/boot pumps)
- Patients taking warfarin do not need additional VTE prophylaxis if they are within their therapeutic range and warfarin is continued.
- Patients on full anticoagulant therapy (LMWH or UFH) or dibigatran, rivaroxaban or other new oral anticoagulants (NOAC), do not need additional VTE prophylaxis.
- For the purposes of critical care it has been decided that a GFR <30 represents the level of renal impairment at which alterations in LMWH prescribing should be considered.

Key Points

Bleeding

The following patients are at increased risk of bleeding and should initially be offered non-pharmacological prophylaxis:

- Active bleeding
- Acquired bleeding disorders (e.g. acute liver failure)
- Concurrent use of warfarin with INR > 2
- Lumbar puncture / epidural / spinal in previous 4 hour or expected in next 12 hours
- Acute stroke (including ICH and SAH)
- Thrombocytopoenia < 75 x 10^9/l
- Uncontrolled systolic hypertension (>230/120 mmHg)
- Untreated inherited bleeding disorder (e.g. Haemophilia)

Graduated Compression Stockings

The following patients are NOT suitable for compression stockings:

- Patients who are sedated and therefore unable to express if their stockings are painful
- Peripheral arterial disease or peripheral arterial bypass e.g. fem-pop etc. These patients may not be suitable for calf/boot pumps either.
- Peripheral neuropathy or other causes of sensory impairment e.g. sedated, epidural etc.
- Fragile 'tissue paper' skin, dermatitis, gangrene or recent skin graft
- Heart failure
- Severe leg oedema or pulmonary oedema from CCF
- Unusual leg size or shape
- Major limb deformity
- Acute Stroke

Compression stockings should be removed **at least** daily for hygiene purposes and to inspect skin. They should be discontinued if there is marking, blistering or discolouration of skin and replaced with foot pumps.

Elective surgical admissions:

If conscious and no epidural:

- If had TEDs fitted pre op – continue with these until mobile.
- If risk of major bleeding low **add** LMWH 6-12 hours post-surgery (unless surgical instruction)

- If not had TEDs fitted and no contraindication – measure and fit and continue with these until mobile.

If sedated, have epidural in situ or other contraindication to Compression Stockings:

- Foot pumps until mobile, unless have peripheral vascular disease.
- If risk of major bleeding low **add** LMWH 6-12 hours post- surgery i.e. Foot pumps **and** LMWH until mobile

Patients who have had spinal surgery should not have LMWH until the surgeons have expressly permitted it (usually >72 hours post-surgery)

All Unplanned Surgical Admissions to ICU

Foot pumps, in the absence of peripheral vascular disease, until mobile.
If risk of major bleeding low (discuss with surgeon who performed operation), **add** LMWH

All Medical Patients Admitted to Critical Care

- If no contraindication to pharmacological prophylaxis, give LMWH
- If pharmacological prophylaxis is contraindicated, place foot pumps until mobile, in the absence of peripheral vascular disease, and reassess need for LMWH daily
- If admitted with **stroke,** place foot pumps, if no peripheral vascular disease,.
- Once risk of haemorrhagic transformation or other bleeding is low, start LMWH. Seek guidance from stroke physicians.

Obesity

Obese patients are at increased risk of VTE. NICE guidance (2010) does not make any specific recommendations regarding increased doses of LMWH in obese patients. However, in light of a paucity of evidence, and the fact that these patients are all high risk for VTE, the available evidence would point towards higher LMWH dosing in this group.

A suggested weight based regime would be:

Weight	Dalteparin dose
<50kg	2500 units daily
50-100kg	5000 units daily
100-150kg	5000 units BD
>150kg	7500 units BD

Ref: HAT Committee of the UK Clinical Pharmacy Association April 2010

Temporary Vena Cava Filters
There are no RCTs investigating vena cava filters but consider offering temporary inferior vena cava filters to patients who are at very high risk of VTE (e.g. patients with a previous VTE event within 6 weeks or an active malignancy or pelvic trauma with active bleeding) and for whom mechanical and pharmacological VTE prophylaxis are contraindicated.

PUBLICATION DETAILS

Author of Clinical Guideline	Rebecca Appelboam, Consultant ICM
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Contact details	x2424, rebeccaappelboam@nhs.net
Version number	1.1
Replaces version number	1.0
Date written	00/06/15
Approving body and date approved	Intensive Care Unit Governance Group: 18/07/18
Review date	18/02/23 (3-6 months prior to expiry date)
Expiry date	18/07/23
Date document becomes live	16/08/18